Biomedical Materials:
Biological Environment and Material Interactions

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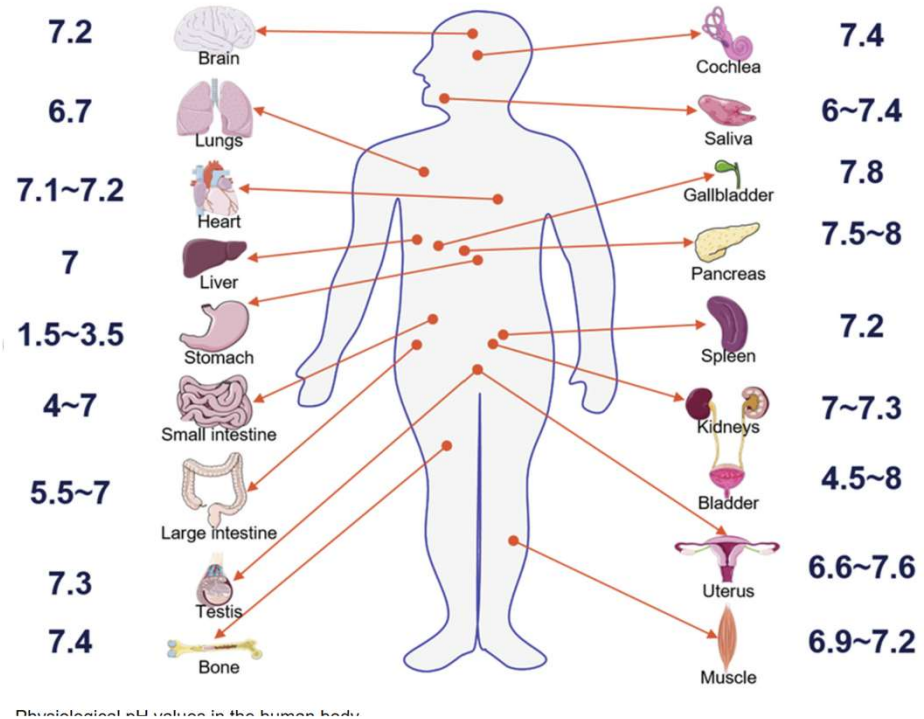
Objective

Understand the environment encountered by a biomaterial and the initial interactions at its surface.

The human body as a materials environment

A metal exposed to seawater may corrode. A polymer exposed to solvents may swell. A ceramic exposed to an aqueous medium may dissolve.

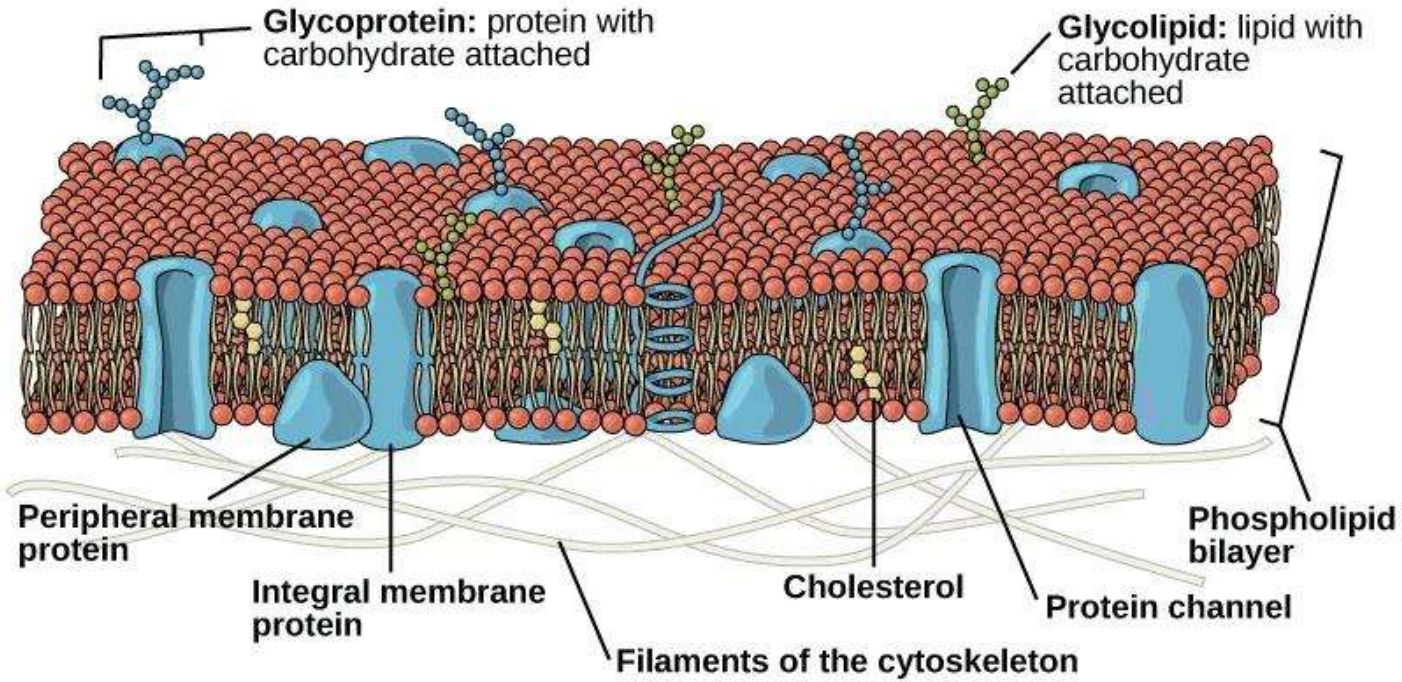
The human body is also an aqueous, chemically reactive environment, but it contains proteins, cells, enzymes, and an active immune system. Physiological pH is regulated but may differ locally.



Essential cell biology for materials engineers

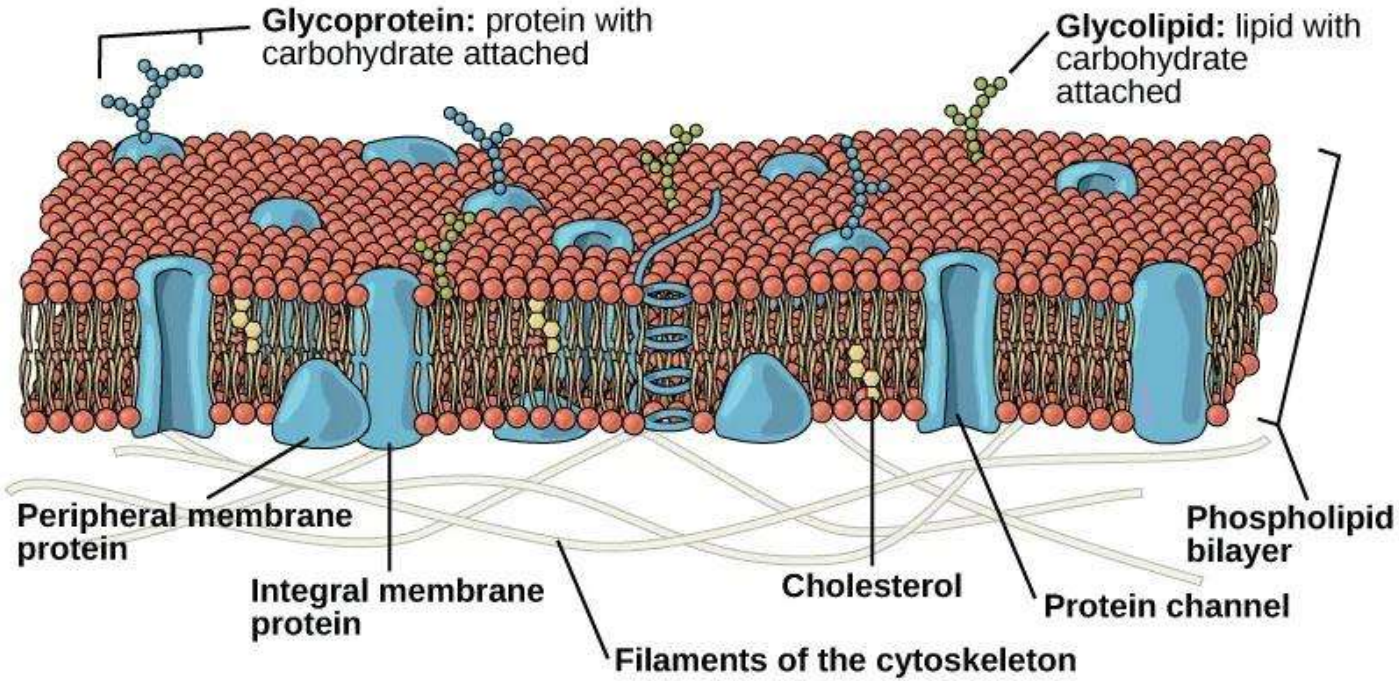
Cell membrane: The cell membrane is a thin, selectively permeable structure that separates the interior of a cell from its surrounding environment. It consists primarily of a phospholipid bilayer containing proteins, cholesterol, and other lipid components. The membrane is typically 5–10 nm thick.

Phospholipids are amphiphilic molecules consisting of a hydrophilic head group and hydrophobic hydrocarbon tails. In an aqueous environment, these molecules spontaneously arrange into a bilayer in which the hydrophobic tails face one another, while the hydrophilic heads are exposed to the intracellular and extracellular fluids.



Essential cell biology

Cell adhesion to an implanted material is generally mediated by membrane receptors that recognize proteins adsorbed onto the material surface. Consequently, the surface properties of a biomaterial can influence cell attachment, spreading, and subsequent cellular responses.

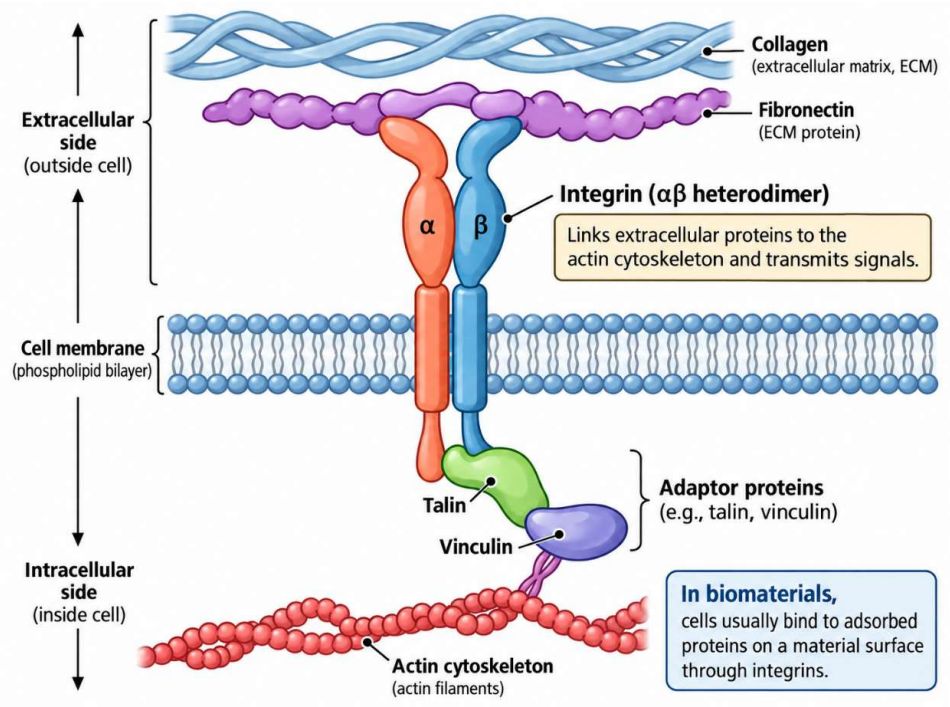


Cell Adhesion Receptors: Integrins

Integrins are located in the cell membrane. they are transmembrane proteins, meaning that they extend through the phospholipid bilayer.

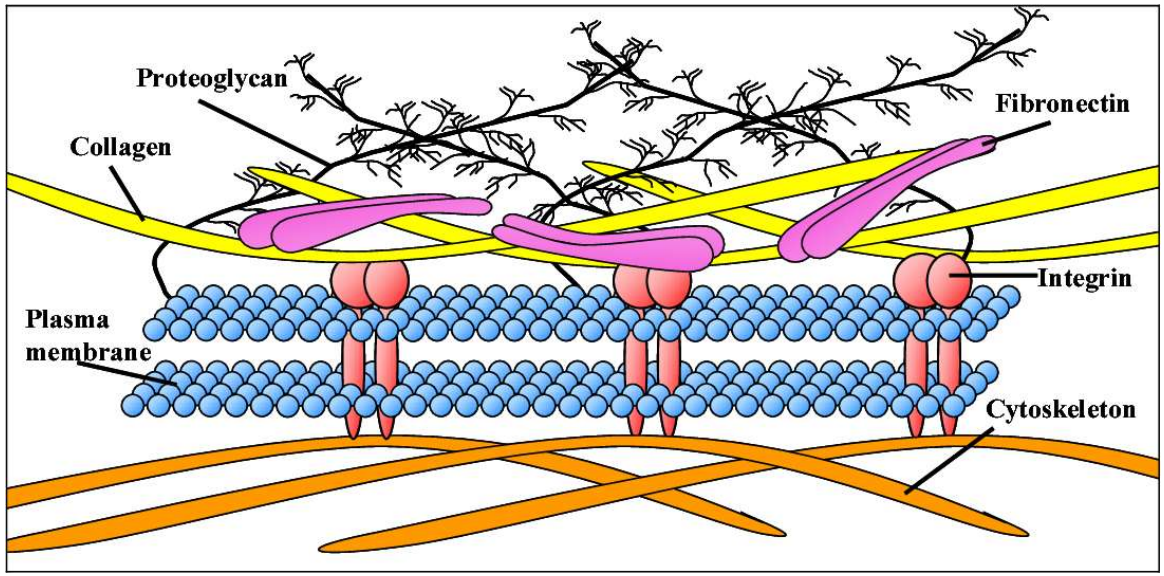
Integrins extend outside the cell and binds to extracellular matrix proteins such as fibronectin, vitronectin, or collagen.

Integrins extend across the cell membrane and establish a connection between extracellular proteins and the **cytoskeleton**.



Fibronectin

Fibronectin is a large, sticky glycoprotein (a protein with attached sugar molecules) found in the extracellular matrix and in blood plasma. It acts as a molecular bridge that connects cells to their surrounding environment. Namely, t contains specific binding sites that allow it to interact with cell membrane receptors known as integrins.

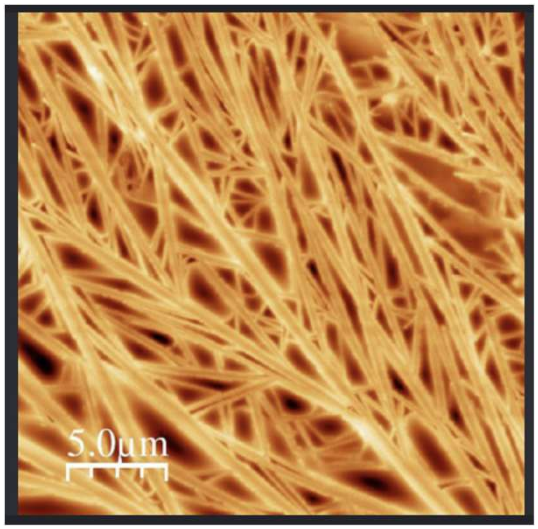


Integrins extend across the cell membrane and establish a connection between extracellular proteins (such as fibronectin) and the **cytoskeleton**.

Extracellular matrix

The ECM is particularly important in tissue engineering and implant integration.

The extracellular matrix (ECM) is a three-dimensional network of proteins and other macromolecules that surrounds cells and provides structural support to tissues.

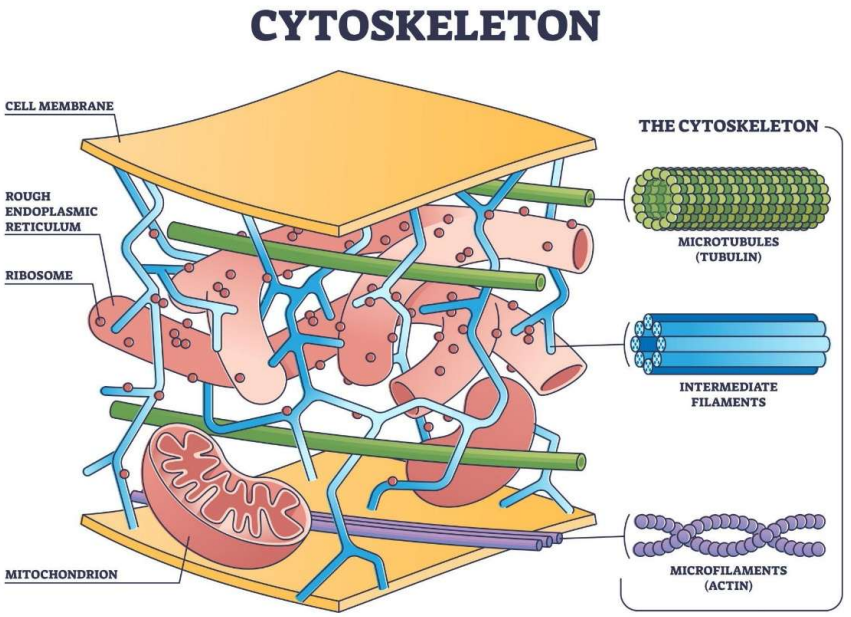


It is a dynamic biological environment whose chemical composition, architecture, and mechanical properties influence cell behavior.

Cytoskeleton

The cytoskeleton is a three-dimensional network of protein filaments extending throughout the cytoplasm of a cell.

Actin filaments (one type of cytoskeleton), one of the main components of the cytoskeleton, are linked to integrins through intracellular adaptor proteins. This connection allows cells to attach to their surroundings and respond to mechanical and biochemical signals.



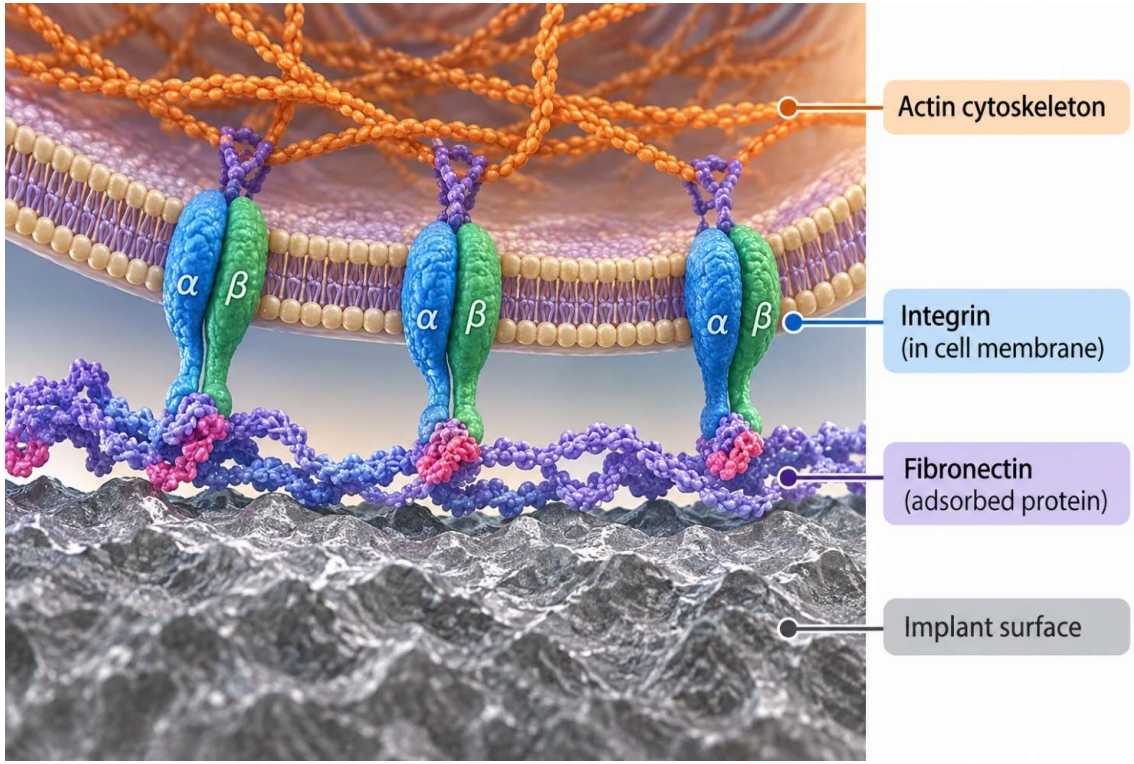
Understanding how cells attach to an implant

1.When an implant is implanted into the body, its surface is rapidly exposed to biological fluids containing proteins.

2. Some of these proteins, including fibronectin (Vitronectin, collagen, and laminin can also adsorb), adsorb onto the metal surface. (The proteins involved depend on the surrounding biological environment, cell type, and material surface.)

3. The adsorbed fibronectin can then provide binding sites for integrins located in the membranes of nearby cells.

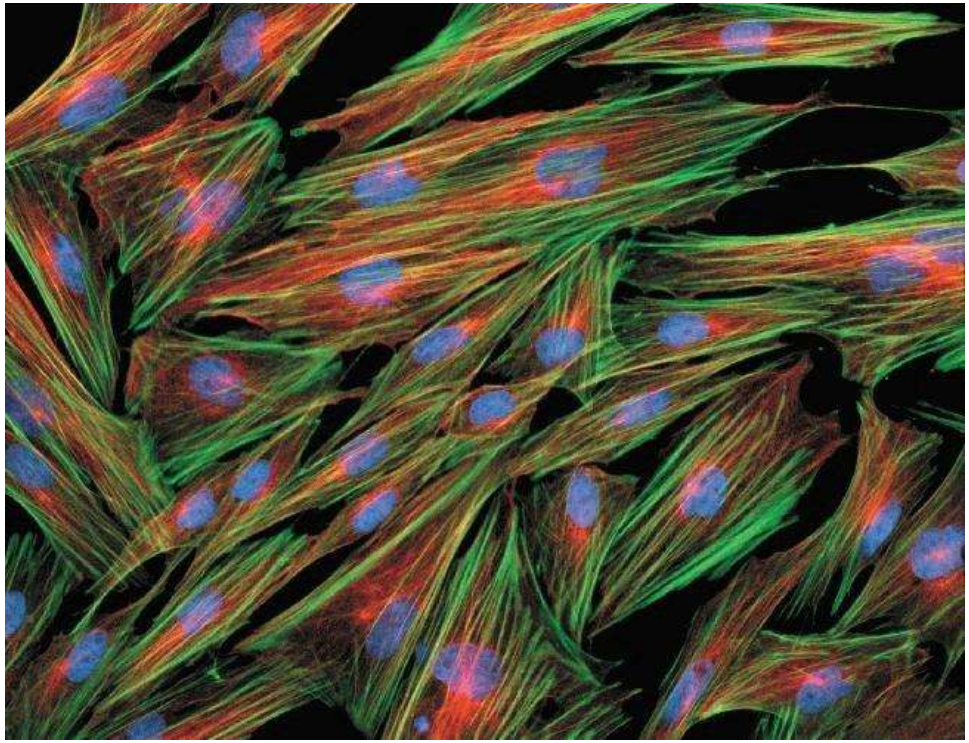
4. The cell does not simply remain attached to the surface through physical adsorption. Instead, integrin-mediated adhesion initiates biochemical signaling and establishes mechanical connections with the cytoskeleton.



Note: α (alpha) and β (beta) represent the two protein subunits that together form an integrin receptor.

Cells: Fibroblasts

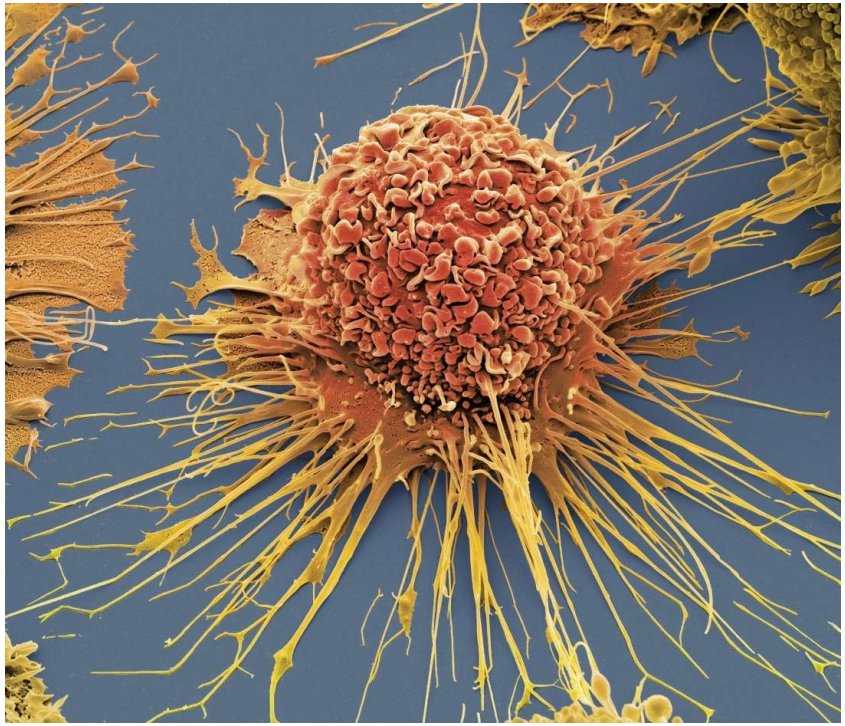
Fibroblasts are connective tissue cells responsible for producing and maintaining many components of the extracellular matrix. Fibroblasts synthesize structural proteins such as **collagen and fibronectin**.



Relevance to biomaterials: Fibroblasts can form connective tissue around an implant. This response may be beneficial during normal tissue repair, but excessive collagen deposition can lead to the formation of a fibrous capsule around the material.

Cells: Macrophages

Macrophages are immune cells that participate in the body's defense against foreign substances, microorganisms, and damaged tissue.

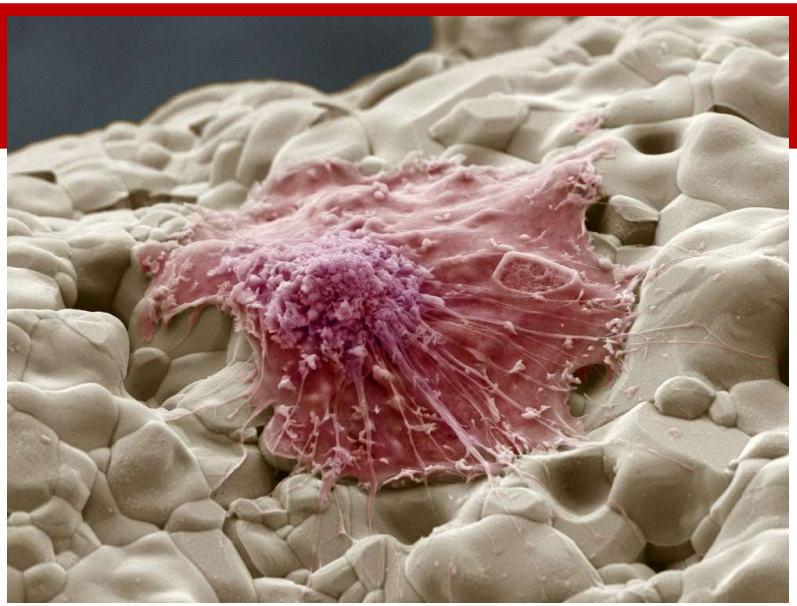


Relevance to biomaterials: Macrophages play a central role in the foreign body reaction and can influence whether an implant becomes integrated into surrounding tissue or develops persistent inflammation and fibrous encapsulation.

regulate inflammation and healing.

Cells: Osteoblast

Osteoblast (a bone-forming cell)



Their primary function is to synthesize the organic component of bone matrix, known as osteoid, which consists mainly of type I collagen and associated non-collagenous proteins.
Osteoblasts also participate in the mineralization of this matrix.



Relevance to biomaterials: Osteoblast attachment, proliferation, differentiation, and matrix mineralization are important in the integration of bone-contacting implants.

https://micro.org.au/inner-space/bone-forming-cell/

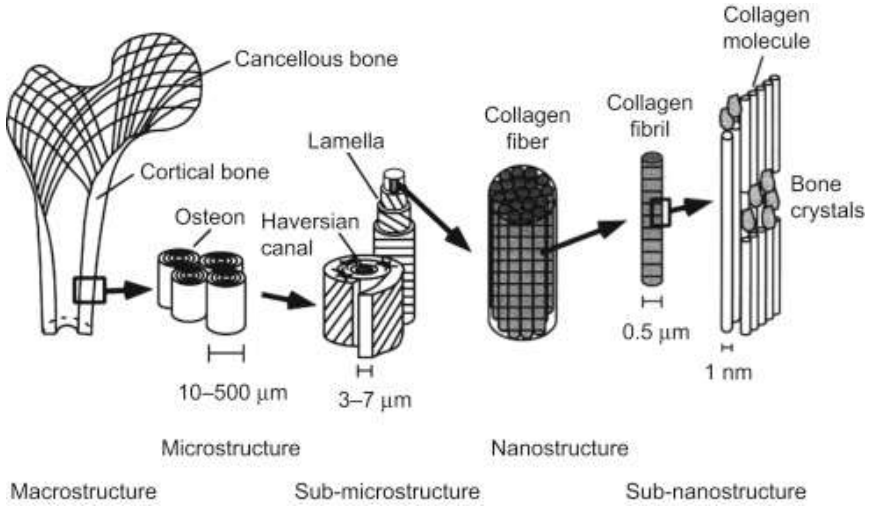
Tissues: Bone Tissue: Structure, Composition, and Properties

Nature of bone (a hierarchical composite structure) is important in biomedical materials engineering. Because we mimic the properties with implants

Bone is a specialized connective tissue consisting of an organic matrix reinforced by an inorganic mineral phase. It provides mechanical support, protects internal organs, facilitates movement, and acts as a reservoir for calcium and phosphate.

A review of bone composition: 60 wt% inorganic material, 25 wt% organic constituents, and 9.7 wt% water.

The inorganic phase consists predominantly of **carbonated apatite (**biological apatite**)**, while the organic matrix is mainly composed of **type I collagen.**



Osteoblasts form new bone matrix, osteoclasts resorb existing bone, and osteocytes participate in mechanosensing and regulation of bone remodeling.

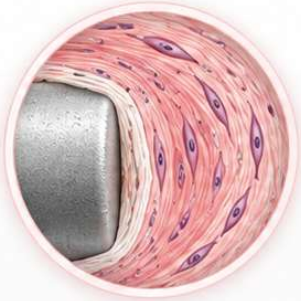
https://www.sciencedirect.com/topics/medicine-and-dentistry/bone-structure

Biocompatibility and Host Response

The host response refers to the biological reactions that occur when a material is introduced into the body.

Possible outcomes of the host response

1. Fibrous encapsulation



Formation of a collagen-rich capsule separating the implant from surrounding tissue

2. Tissue integration



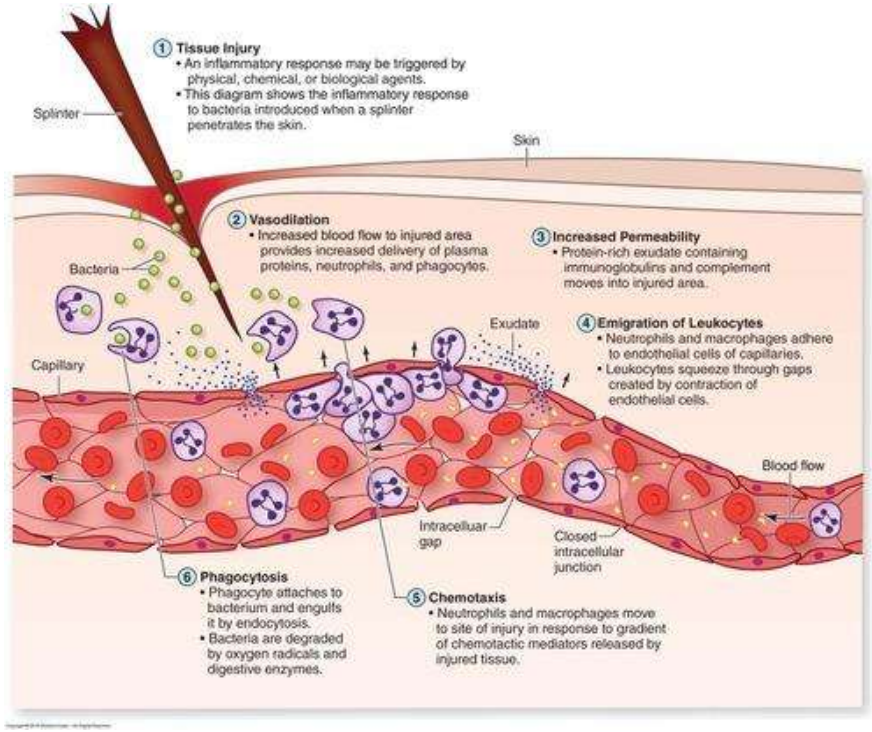
Direct integration with surrounding tissue, such as osseointegration

For bone-contacting implants, osseointegration is generally desirable, whereas fibrous encapsulation is undesirable because it can interfere with stable bone–implant fixation.

Inflammation Following Implantation

When a biomaterial is implanted, the surgical procedure causes tissue injury and triggers an inflammatory response. The material itself may subsequently influence the intensity and duration of this response.

Acute inflammation begins shortly after implantation and generally lasts from hours to several days. (it is something to do with regulation of tissue repair and it is not bad!)



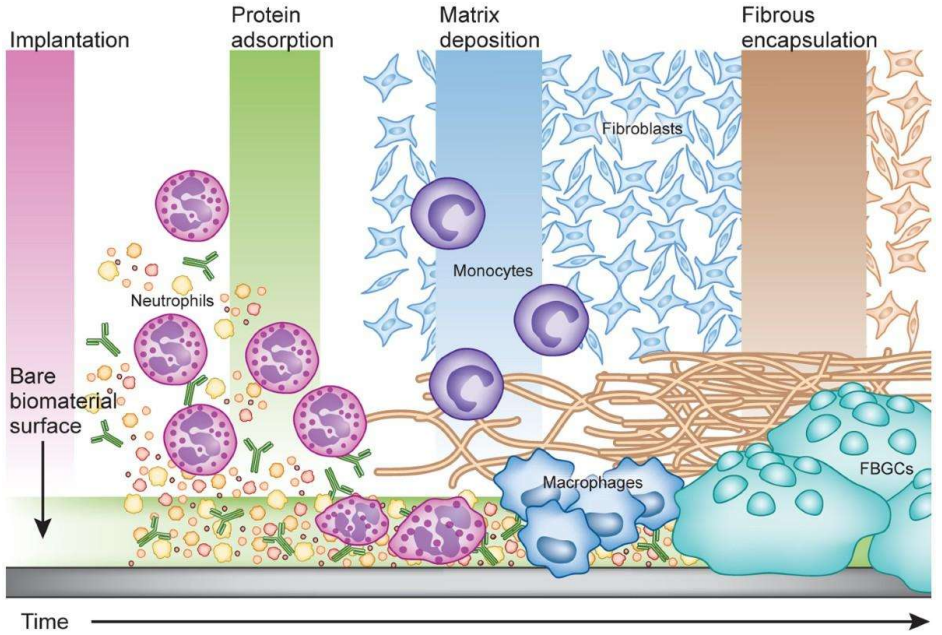
Inflamation: iltihaplanma

Fibrous tissue is composed predominantly of collagen fibers

Inflammation Following Implantation

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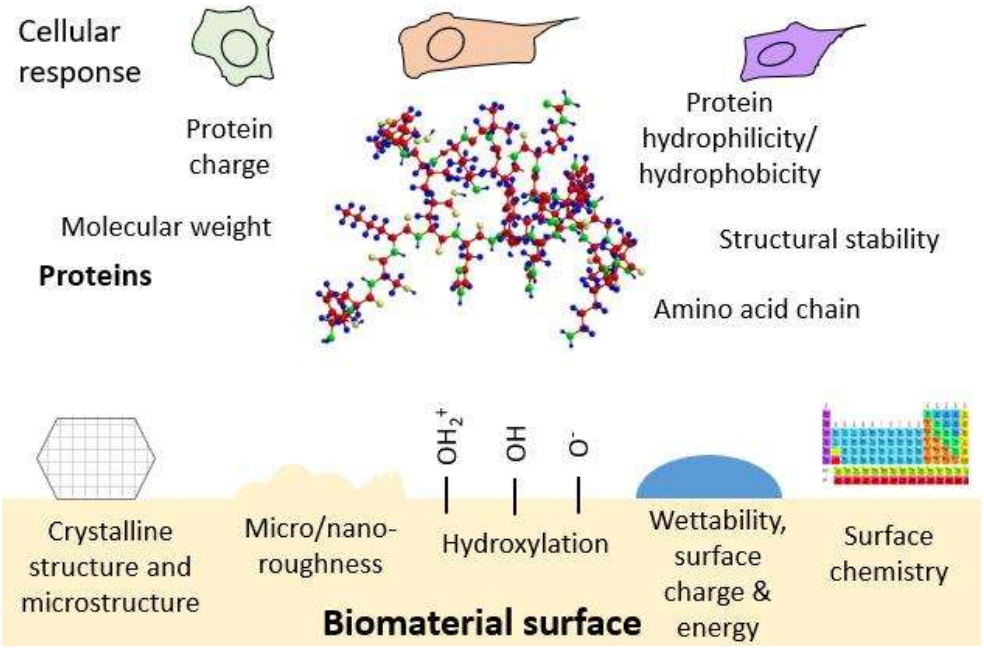
Chronic inflammation may develop when the inflammatory stimulus persists or the normal resolution process is disrupted. Macrophages remain active at the implant interface. Persistent inflammation may interfere with tissue integration and implant function. End up bad by fibrous tissue formation around the implant



a thick fibrous layer surrounding a bone-contacting implant may prevent direct bone–implant contact.

Protein adsorption: Implant surface chemistry

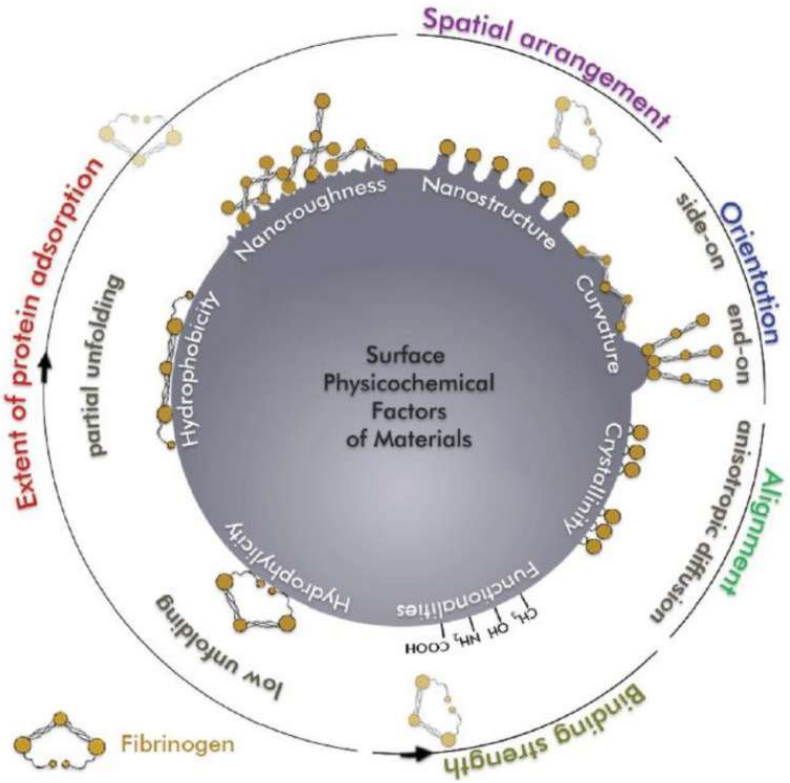
Protein adsorption is governed by physical and chemical interactions between proteins and the biomaterial surface.
Albumin, fibrinogen, fibronectin, and vitronectin are among the proteins commonly found on implanted material surfaces.
The composition, orientation, and conformation of adsorbed proteins affect subsequent cell attachment and biological responses.



Surface chemistry, wettability, charge, crystallinity, and micro/nanoscale roughness

Factors Influencing Protein Adsorption

Surface property	Brief explanation
Surface chemistry	The chemical composition and functional groups on the surface determine how proteins interact with the material.
Wettability	Describes how readily water spreads over a surface. It affects interfacial hydration and protein adsorption.
Surface charge	Positively or negatively charged surfaces can attract or repel charged regions of proteins through electrostatic interactions.
Crystallinity	The arrangement of atoms in a material can influence its surface structure, energy, and available protein-binding sites.
Micro/nanoscale roughness	Surface irregularities change the available area and local geometry for protein adsorption.



Requirements for a successful implant surface

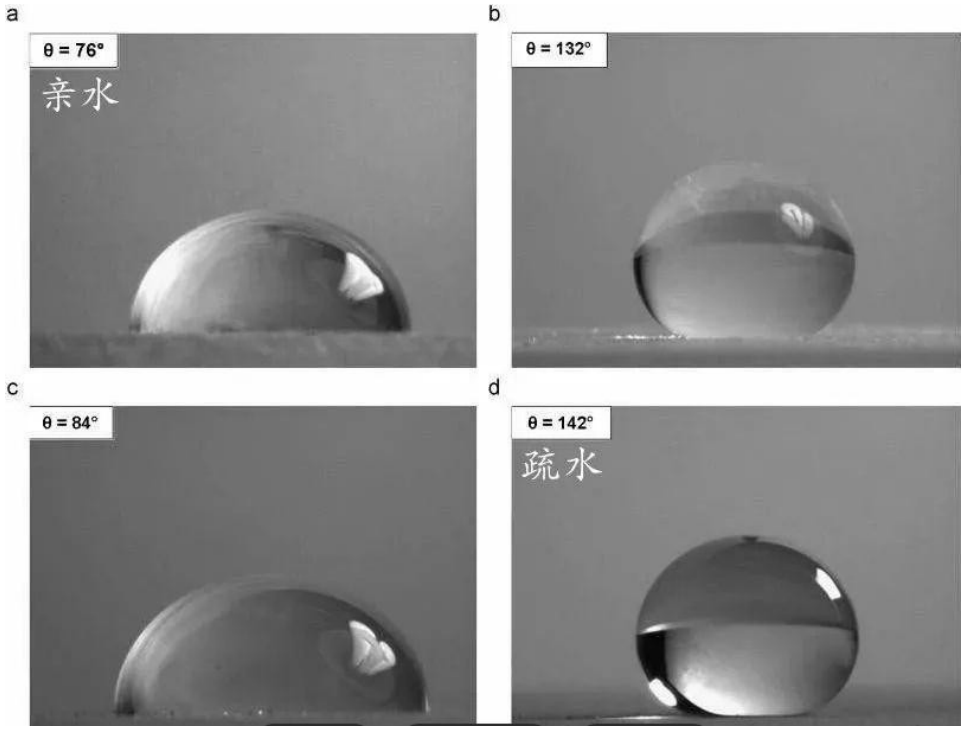
There is no no universally ideal combination of hydrophilicity, surface charge, and roughness.

A catheter may require low friction, minimal bacterial adhesion, and reduced protein fouling.

A bone implant should generally promote osteoblast attachment.

A blood-contacting implant should generally minimize unwanted protein adsorption.

Hydrophilic or hydrophobic?



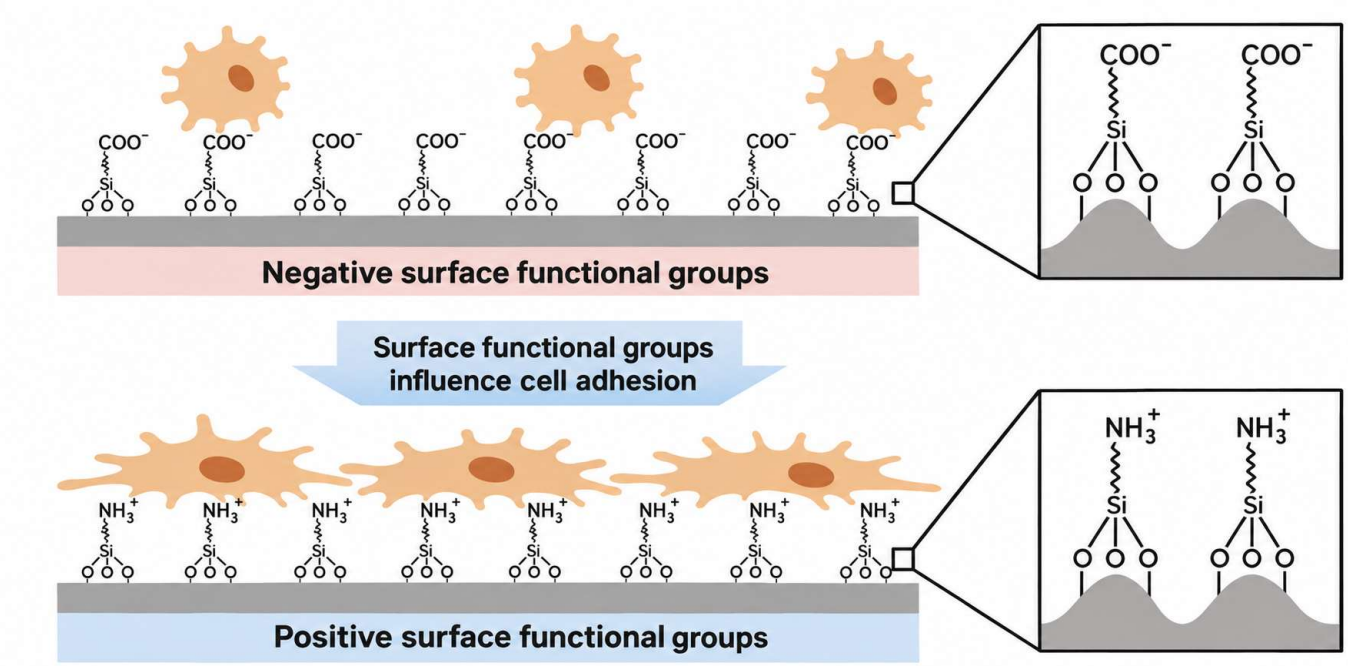
Hydrophilic surface support favorable protein interactions.

Hydrophobic interactions promote adsorption of proteins through hydrophobic interactions and can cause changes in their conformation.

For titanium dental and orthopedic implants, hydrophilic surfaces are often desirable

Surface charge

They can influence calcium ion interactions and calcium phosphate nucleation, depending on the surface chemistry and solution conditions. This can be beneficial for bone-facing materials. However, negative charge alone does not guarantee osseointegration.



https://doi.org/10.1039/c6nr00649c

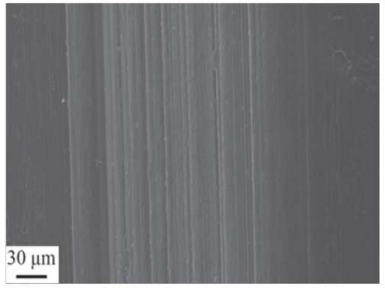
Surface roughness

Smooth surface

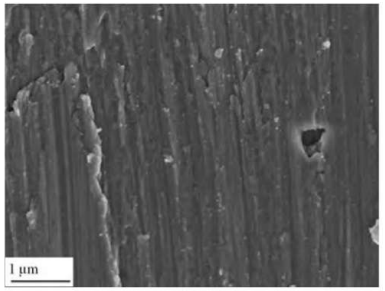
May be favorable where low friction or limited adhesion is desired.



(a)



(b)

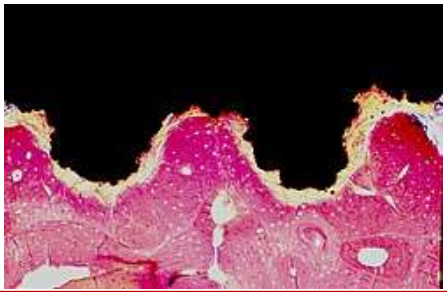


(c)

Roughened surface

Controlled microroughness is often beneficial for dental and orthopedic implants because it can support bone apposition and mechanical interlocking.





Titanium implant (black) **integrated** into bone (red) (Wikipedia)

Osseointegration is the direct structural and functional connection between living bone and the surface of a load-bearing artificial implant

Titanium—When a Corrosion-Resistant Metal Fails

•The SR-71 was constructed largely from titanium alloys because of their excellent strength-to-weight ratio and elevated-temperature performance.
•During manufacturing, spot-welded titanium wing panels produced in summer experienced premature failures.
•Surprisingly, panels manufactured during winter did not exhibit the same problem.



Engineers traced the failures to chlorine contamination from the municipal water used to wash components.

Metallic biomedical materials

Biocompatibility and long-term material performance depend not only on the intrinsic properties of a material, but also on its surface condition, processing history, and interactions with the surrounding environment.





If titanium can experience corrosion-related failure during aircraft manufacturing, can we assume that a titanium implant will remain completely stable inside the human body?

Questions

Why can two biomaterials with the same bulk composition but different surface treatments produce different cellular responses?

We have two titanium implants: one hydrophobic and smooth, and the other hydrophilic and moderately roughened. Which would be more suitable for osseointegration, and why?